



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Grocery store management system

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Agenda

- ❖ Objective of the Project.
- ❖ Dataset Overview.
- ❖ ER Diagram and schema explanation.
- ❖ Key analysis questions.
- ❖ SQL query results with screenshots or summaries.
- ❖ Final business insights.
- ❖ Challenges working on SQL – Data Analysis Project.
- ❖ Conclusion.

Business Problem :

- ❑ The grocery store lacks a structured system to manage products, suppliers, customers, and orders efficiently.
- ❑ Sales data is not properly analyzed to identify revenue trends and high-performing products.
- ❑ There is no clear tracking of employee performance and supplier contribution.
- ❑ Business decisions are made without proper data insights.
- ❑ Manual data handling can lead to errors and data inconsistency.

Business Problem Statement :

- Maintain proper inventory tracking to avoid stock shortages and expired products.
- Use a computerized billing system to reduce manual errors and save time.
- Monitor daily sales and profit reports to identify fast-moving and slow moving items.
- Control expenses and analyze data regularly to improve overall business growth.

Objectives :

The main goals of this SQL project are:

- ❖ **To design and implement a relational database for a grocery store**
- ❖ **To retrieve and manipulate data using SQL queries**
- ❖ **To perform data analysis to identify top customers, best-selling products, and revenue trends**
- ❖ **To analyze supplier contribution, employee performance, and order behavior**
- ❖ **To practice advanced SQL concepts like JOINS, GROUP BY, Aggregations, Subqueries, and Window functions**

Dataset Overview

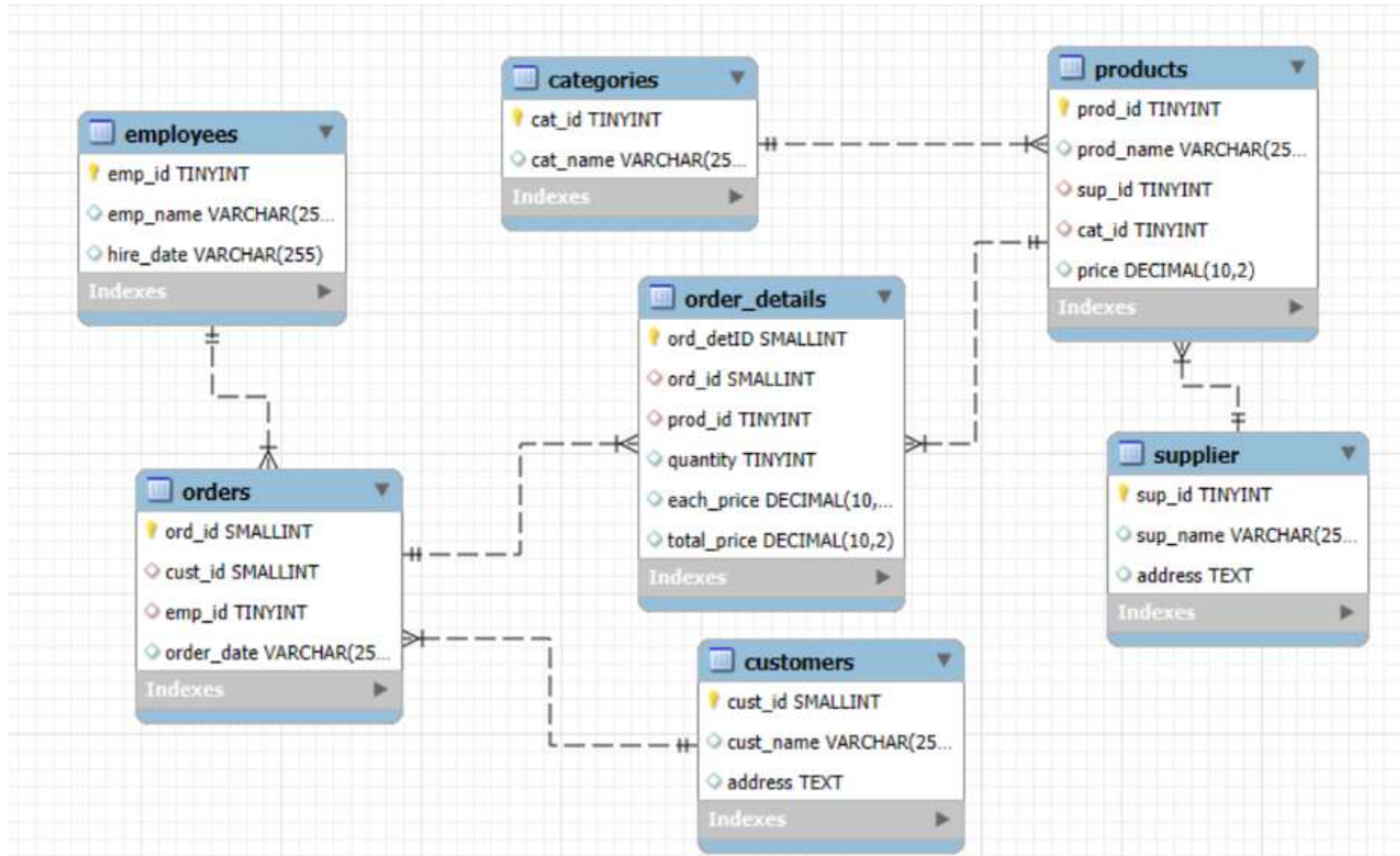
The dataset consists of 7 main tables:

- ☐ Products
- ☐ Suppliers
- ☐ Employees
- ☐ Orders
- ☐ Order Details
- ☐ Customers
- ☐ Categories

These tables represent real grocery store sales operations and are connected using primary and foreign keys.

ER – Diagram

The ER diagram shows the relationships between all tables. Most relationships are One-to-Many



Schema Explanation :

- Each Supplier acts as a master record and is connected to the Products table through the foreign key (sup_id).
- Each Category is linked to multiple Products using (cat_id).
- Each Customer is connected to multiple Orders using (cust_id).
- Each Employee handles multiple Orders through (emp_id).
- Each Order is connected to multiple products through the Order_Details table using (ord_id).
- Each Product can appear in multiple Order_Details records using (prod_id).

This design creates one-to-many relationships where:

One Supplier → Many Products

One Category → Many Products

One Customer → Many Orders

One Employee → Many Orders

One Order → Many Order_Details

One Product → Many Order_Details

This schema design ensures proper data organization, maintains relationships using foreign keys, and prevents orphan records using cascade constraints

Cascade Rules in Grocery Store Database

ON UPDATE CASCADE:

- It automatically updates foreign key values in related tables when the primary key changes.
- This ensures data consistency across related tables and prevents broken relationships.

For example, if a supplier ID changes, all related products will automatically update.

Where used in this project:

- supplier → products
- categories → products
- customers → orders
- employees → orders

ON DELETE CASCADE :

- It automatically deletes related child records when a parent record is removed.
- Keeps the database clean and valid when deleting orders/customers/suppliers.
- This prevents orphan records and maintains referential integrity.

For example, when an order is deleted, the related order details are also deleted automatically.

Where used in this project:

- orders → order_details
- supplier → products
- categories → product

Update cascade and Delete cascade

-- 7. Order_Details Table

- ```
CREATE TABLE IF NOT EXISTS order_details (
 ord_detID SMALLINT AUTO_INCREMENT PRIMARY KEY,
 ord_id SMALLINT,
 prod_id TINYINT,
 quantity TINYINT,
 each_price DECIMAL(10,2),
 total_price DECIMAL(10,2),
 FOREIGN KEY (ord_id) REFERENCES orders(ord_id)
 ON UPDATE CASCADE ON DELETE CASCADE,
 FOREIGN KEY (prod_id) REFERENCES products(prod_id)
 ON UPDATE CASCADE ON DELETE CASCADE
);
```

# 1. Customer Insights

```
-- Which customers have placed the highest number of orders?
select o.cust_id,c.cust_name, count(*) as total_cust
from orders o
join customers c on
o.cust_id=c.cust_id
group by cust_id
order by total_cust desc;
```

```
-- Who are the top 5 customers by total purchase amount?
select o.cust_id,c.cust_name, sum(total_price) as total_
from orders o
join customers c
on c.cust_id=o.cust_id
join order_details od
on o.ord_id=od.ord_id
group by o.cust_id
order by total_pur desc
limit 6;
```

## 2. Product Performance

-- How many products exist in each category?

```
select * from categories;
```

```
select * from products;
```

```
select p.cat_id, count(p.prod_name) as each_product
from products p
group by p.cat_id;
```

-- What is the average price of products by category?

```
select p.cat_id , c.cat_name, avg(price) as avg_prz
from products p
join categories c
on p.cat_id=c.cat_id
group by cat_id;
```

### 3. Sales and Order Trends

```
-- What is the average value per order?
select ord_id, avg(total_price) as avg_value
from order_details
group by ord_id;
```

```
-- On which dates were the most orders placed?
select order_date, count(ord_id) as most_orders
from orders
group by order_date
having count(ord_id)=(
select max(order_count)
from(select count(ord_id) as order_count
from orders
group by order_date)t
);
```



## 4. Supplier Contribution

-- Which supplier provides the most products?

```
select s.sup_name, p.sup_id, count(p.prod_id) as high_supplier
from products p
join supplier s on
s.sup_id=p.sup_id
group by p.sup_id
order by high_supplier desc
limit 1;
```

-- What is the average price of products from each supplier?

```
select s.sup_name, p.sup_id, avg(p.price) as sup_avg_price
from products p
join supplier s on
s.sup_id=p.sup_id
group by p.sup_id;
```

## 5. Employee Performance

-- Which employees have handled the most orders?

```
select o.emp_id ,e.emp_name, count(o.ord_id) as most_orders
from orders o
join employees e on
o.emp_id=e.emp_id
group by emp_id
order by most_orders desc
limit 1;
```

-- What is the total sales value processed by each employee?

```
select o.emp_id ,e.emp_name, sum(od.total_price) as total_sales
from orders o
join employees e on
e.emp_id=o.emp_id
join order_details od on
o.ord_id=od.ord_id
group by o.emp_id
order by total_sales desc;
```

## 6. Order Details Deep Dive

-- What is the average quantity ordered per product?

```
select od.prod_id,p.prod_name, avg(od.quantity) as avg_quant
from order_details od
join products p on
od.prod_id=p.prod_id
group by od.prod_id;
```

-- How does the unit price vary across products and orders?

```
SELECT od.prod_id,p.prod_name,
 AVG(each_price) AS avg_unit_price,
 MIN(each_price) AS min_unit_price,
 MAX(each_price) AS max_unit_price
FROM order_details od
join products p on
od.prod_id=p.prod_id
GROUP BY od.prod_id;
```



# Combined Business Insights :

## Supplier Insights :

- One supplier provides the highest number of products in the store.
- The same supplier generates the highest revenue compared to others.
- The business depends more on this major supplier for overall sales performance.

## Employee Insights :

- Some employees handle more orders than others.
- Employees who process more orders generate higher sales.
- Balanced workload and proper training can improve employee performance.

## Order Insights :

- Higher quantity orders generate higher total revenue.
- Some products are ordered more frequently than others.
- There are small price differences for the same product in different orders.

# Challenges Faced :

**During this project, I faced challenges such as:**

- ❖ **Understanding table relationships.**
- ❖ **Writing complex JOIN queries.**
- ❖ **Performing accurate aggregations.**
- ❖ **Interpreting query results correctly.**
- ❖ **These challenges helped me improve my SQL skills.**

# Conclusion :

- ❖ In conclusion, this project demonstrates how SQL can be used to analyze real-world business data.
- ❖ SQL helped extract actionable business insights
- ❖ Project demonstrates capability in data interpretation.
- ❖ It improved my database design skills and analytical thinking.
- ❖ The analysis provides valuable insights to improve supplier management, employee performance, and sales operations.
- ❖ Sets foundation for advanced analytics and dashboards

THANK  
YOU

